TÉCNICO LISBOA

COMPUTER GRAPHICS FOR GAMES

P2 2026-27

Carlos Martinho

QUARTER PLANNING

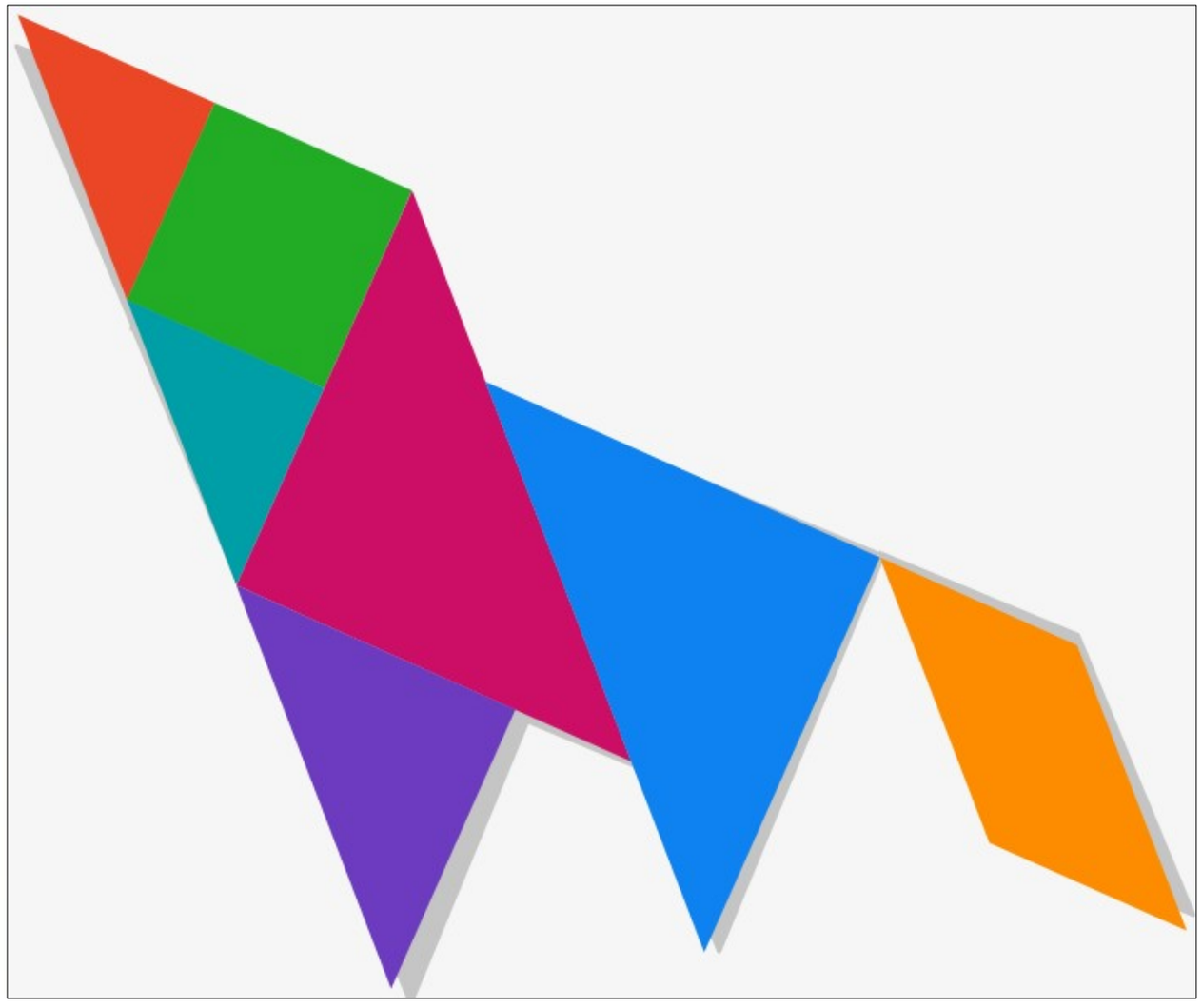


November 2026

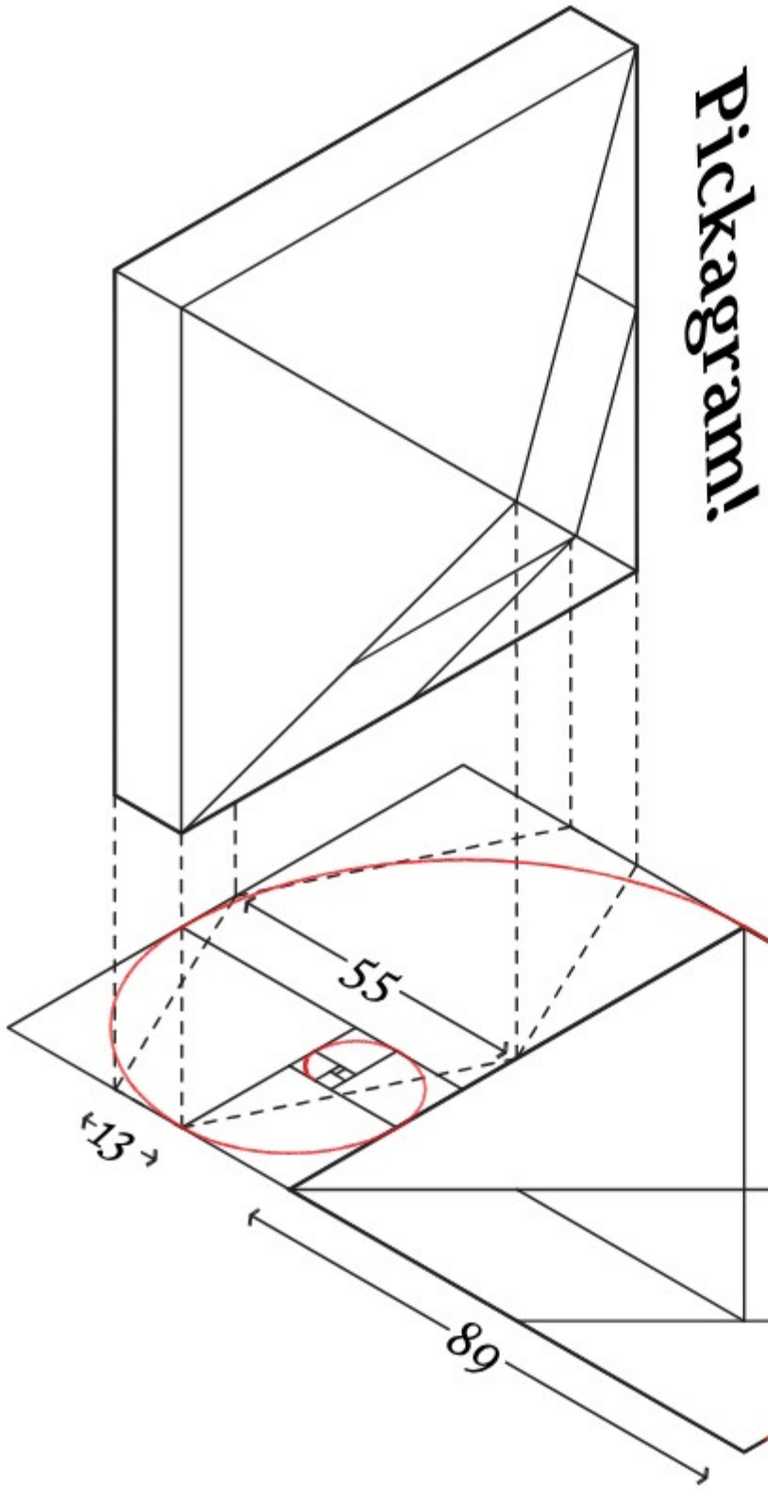
W	Date	Lectures	Labs	Assignments due	Grade
1	Nov 3 (Tue)	Introduction, Review	Team forming C++ IDE		
1	Nov 6 (Fri)	Mathematics for CG (part 1): dot, cross, matrices	GLM / GLM setup		
2	Nov 10 (Tue)	Mathematics for CG (part 2): matrices, quaternions	GLM library	**A1 : GLM (50% in class)** Nov 12 (Thu)	**1.0**
2	Nov 13 (Fri)	Programmable Graphics Pipeline, Drawing in OpenGL	OpenGL / GLFW / GLEW Setup	Tan/Picka-gram Shape (in class)	
3	Nov 17 (Tue)	Viewing Pipeline Viewing in OpenGL	Drawing + Transformations	**A2 : 2D Tangram** Nov 19 (Thu)	**3.0**
3	Nov 20 (Fri)	Geometry Management Meshes and Scene Graphs	Viewing + Quaternions		

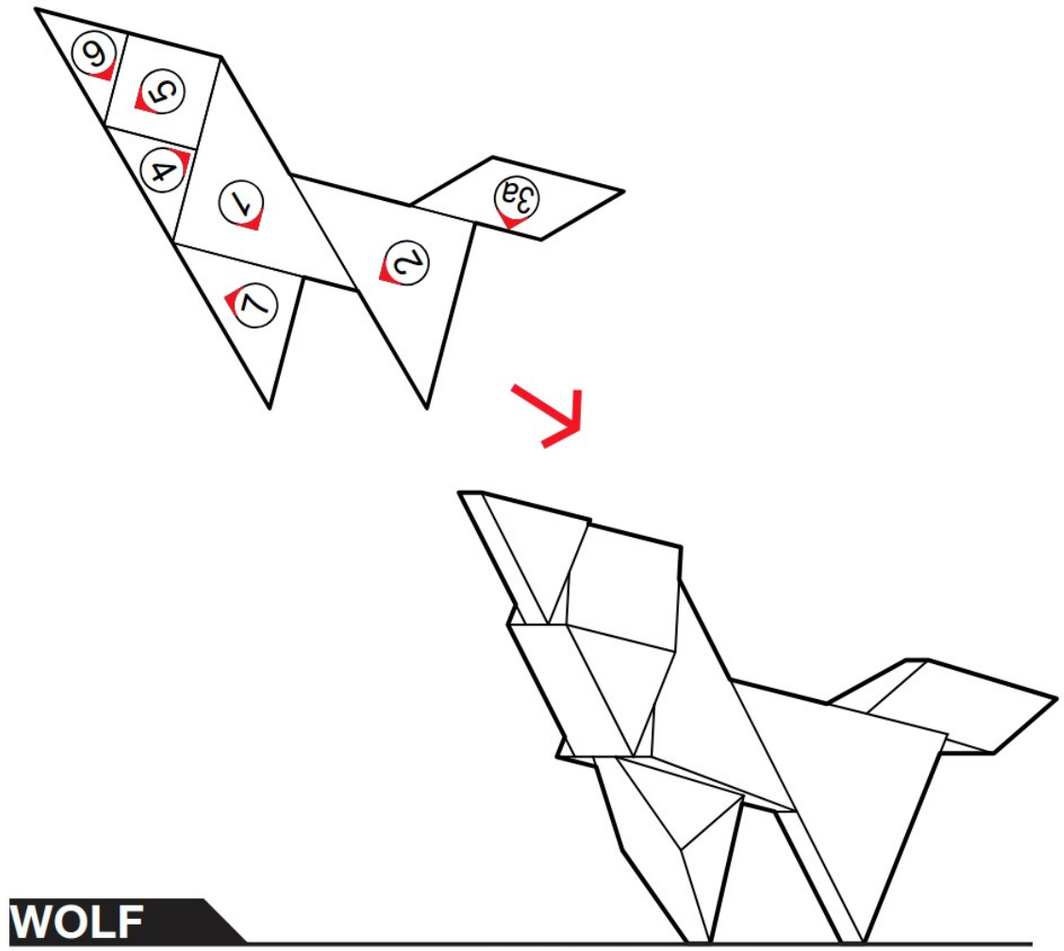
Tangram



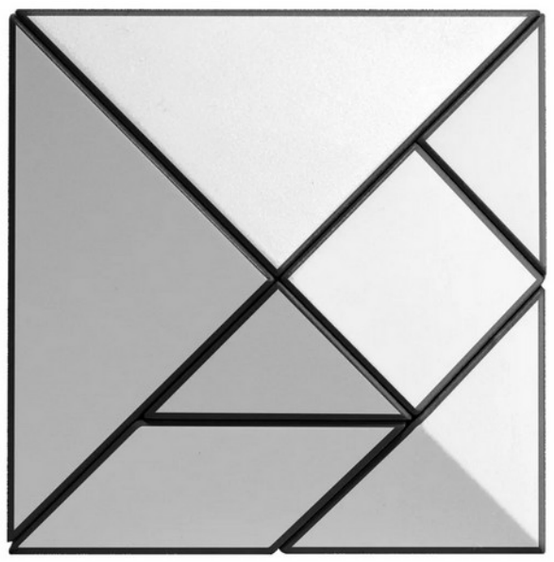


Pickagram



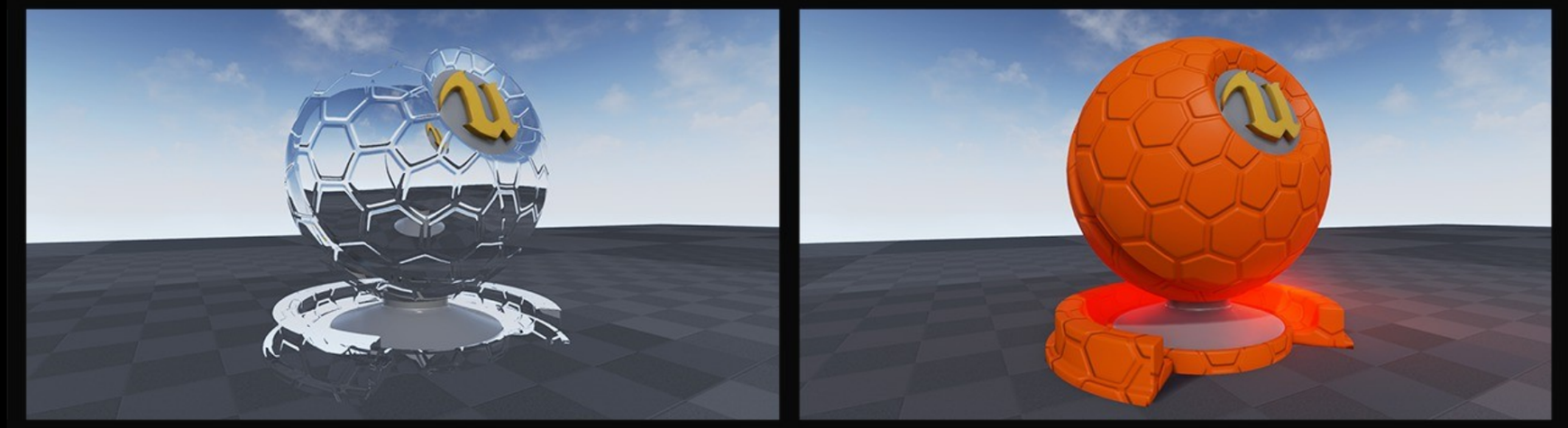


WOLF



November - December 2026

W	Date	Lectures	Labs	Assignment due	Grade
4	Nov 24 (Tue)	GLSL Shaders Lighting (1): Cel Shading	Meshes + ASSIMP Library		
4	Nov 27 (Fri)	Lighting (2): Gouraud and Phong Shading	Project Support	**A3 : 3D Pickagram** Nov 30 (Mon)	**4.0**
5	Dec 1 * (Tue feriado)	Mapping (1): Texturing, Sampling and Filtering	Shader Project Individual Project Definition	**A4 : Project Proposal** Dec 3 (Thu)	**1.0**
5	Dec 4 (Fri)	Mapping (2): Procedural Textures and Noise	Goal Definition and Execution Planning		
6	Dec 8 * (Tue feriado)	Mapping (3): Environment Mapping and Fresnel Reflectance	Project Support		
6	Dec 11 (Fri)	Mapping (4): Render Target Textures, Tangent Space, Normal and Displacement Mapping	Project Support		



December 2026 - January 2027

W	Date	Lectures	Labs	Assignments Due	Grade
7	Dec 15 (Tue)	Mapping (5): Ambient Occlusion and Physically Based Rendering	Project Support		
7	Dec 18 (Fri)	(project report Q&A)	Project Support	**A5 : Shader Project** Jan 4 (Mon)	**8.0**
8				**A5: Project Video Report** Jan 4 (Mon) **Project Discussions** Jan 6-7 (Wed-Thu)	**3.0**

IN BRIEF

- 40% - Base Project (teams of 2, 3 checkpoints)
- 60% - Shader Project (individual, 2 checkpoints)

IMPORTANT

Reports must include:

- Declaration of use of AI in assessment components
- Declaration of honour for group work

Check the *evaluation methods* section of the course website for more information.

CONTACT

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CGJ Office hours on Fénix.